

1. A candle weighs 300 grams (g). Each hour the candle is lit, it burns 4 g of its wax.
- a. Let h be the number of hours the candle burns. Write an equation that can be used to determine the number of hours it takes for the candle to weigh 206.4 g.

Equation	$300 - 4h = 206.4$
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- b. Solve the equation to determine the number of hours it takes for the candle to weigh 206.4 g. Show your work.

Number of hours	23.4
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2. Jayden has two jobs. One week his paycheck was the same for both jobs. As a server he earned \$9.64 per hour plus \$82.35 in tips. Working at a jewelry store he earned \$14.86 per hour plus \$17.10 in sales bonuses. Write an equation that can be used to determine the number of hours, h , that Jayden worked the week his paycheck was the same for both jobs.

Equation	$9.64h + 82.35 = 14.86h + 17.10$
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3. Tim and Jessica plan to rent a camper van for a vacation. The van costs \$183 per day. The rental includes 120 miles for free and then charges \$0.39 per mile.

- a. The total price Tim and Jessica paid to rent the camper van is \$598.74. Write an equation that models the cost of the van rental in terms of miles traveled, m .

Equation	$598.74 = 183 + 0.39(m - 120)$
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- b. Use your equation to determine how many miles Tim and Jessica traveled if they paid \$598.74. Show your work.

Number of miles	1,186
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4. An orange grove manager uses two different sized buckets to load oranges into a truck. The first bucket can transport x pounds of oranges, and the second bucket can transport y pounds of oranges. The first bucket is filled up G times and the second is filled up H times. The total amount of oranges, T , that have been loaded can be found using the equation $T = Gx + Hy$.

Solve the equation $T = Gx + Hy$ for H .

$$H = \frac{T - Gx}{y}$$

5. In the construction of bridges, engineers use a formula to help determine the allowable stress on a bridge rocker, a component of the bridge that handles the stress and weight placed on the pavement of the bridge. The engineers use the formula $F = \frac{11}{300}d(F - 13)$, where F is the strength of the steel in kilopound per square inch and d is the diameter of the rocker in millimeters.

Solve the formula $F = \frac{11}{300}d(F - 13)$ for d .

$$d = \frac{300}{11} \cdot \frac{F}{F - 13}$$

6. Complete the statements.

a. The system $\begin{cases} y - 3 = 12(4 - x) \\ y = 3(-4x + 3) \end{cases}$

- ☒ has no solution.
- ☐ has one solution.
- ☐ has an infinite number of solutions.

b. The system $\begin{cases} y = 10 - 2x \\ 6x + 3y = 30 \end{cases}$

- ☐ has no solution.
- ☐ has one solution.
- ☒ has an infinite number of solutions.

c. The system $\begin{cases} 4x + y = -16 \\ 2x = -5y - 26 \end{cases}$

- ☐ has no solution.
- ☒ has one solution.
- ☐ has an infinite number of solutions.

7. Serena just started working at the local salon. She is paid \$9.85 per hour and earns, on average, \$58.50 in tips each day. Roseline just started working at a bank. Since she does not receive tips, she earns \$16.35 an hour. The system of equations $\begin{cases} y = 9.85x + 58.50 \\ y = 16.35x \end{cases}$ can be used to represent the amount of money earned, y , each day for working x hours. Determine the number of hours Serena and Roseline have to work to earn the same amount of money. Show your work.

Number of hours	9
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8. In many countries, cigarette smoking has decreased in the 21st century. The percentage of smokers in Russia can be represented by the equation $36x + 100y = 346$, where y is the percentage of adults who smoke and x is the number of years since 2000. The percentage of smokers in Poland can be represented by the equation $18x + 50y = 176$, where y is the percentage of adults who smoke and x is the number of years since 2000.

Determine the solution to the system $\begin{cases} 36x + 100y = 346 \\ 18x + 50y = 176 \end{cases}$. Show your work.

Solution	No Solution
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Solve each system of equations. Show your work.

9.	$y = -5x + 17$ $2x + 3y = 12$	
Solution	(3, 2)	

10.	$-2x + y = -18$ $3x + 4y = 5$	
Solution	(7, -4)	

11.	$4x + 9y = -15$ $-5x + 7y = 37$	
Solution	(-6, 1)	

12. Solve the system $\begin{cases} y = \frac{3}{2}x - 3 \\ y = 4x + 2 \end{cases}$ by graphing.

Solution

$(-2, -6)$

